

Classifying machine learning problems

Example 1: predicting house prices

- Type of learning: Supervised learning
- Problem type: Regression
- Y-value: house price
- X-value: house size, bedrooms, location, amenities

Historical data is important because it contains known prices which allows the model to identify similarities, patterns and thus allow the model to predict potential future outcomes.

Examples 2: finding fraudulent transactions

- Type of learning: Supervised learning
- Problem type: classification
- X-values: transaction amount, time, location, and previous transaction patterns
- Y-value: fraudulent or legitimate outcome

The model learns from transactions with a known transactions and predicts new transactions

Example 3: Grouping customers by purchasing behaviour

- Type of learning: Unsupervised learning
- Type of problem: Clustering
- X-values(features): purchase frequency, average spending, customer age, purchase history
- Y value(target): not predefined

The algorithm is used to discover customer profiles without being given any labels beforehand.

2. testing models on unseen data helps determine how the model works with generalized data than just memorizing data it already has.

Test data also provides a more independent assessment of the models predictive performance.

3. end-to-end ML workflow

1. Collect raw data -> 2. Clean and prepare collected data -> 3. Exploratory data analysis
->4. Feature engineering ->5. Split the dataset->6. Choose and train a model ->7.
Evaluate model performance -> 8. Adjust and improve model -> 9. Deploy->10.
Continuously monitor the model